

2 The light-independent reactions of photosynthesis use the products of the light-dependent reactions to produce simple sugars.

- (a) (i) Which row of the table shows the products of the light-dependent reactions that are used in the light-independent reactions?

(1)

	ATP produced by	NADP
☐ A	cyclic photophosphorylation	oxidised
☐ B	cyclic photophosphorylation	reduced
☐ C	non-cyclic photophosphorylation	oxidised
☐ D	non-cyclic photophosphorylation	reduced

- (ii) Simple sugars have the formula $C_nH_{2n}O_n$.

Name the inorganic molecule that provides each element in a simple sugar.

(2)

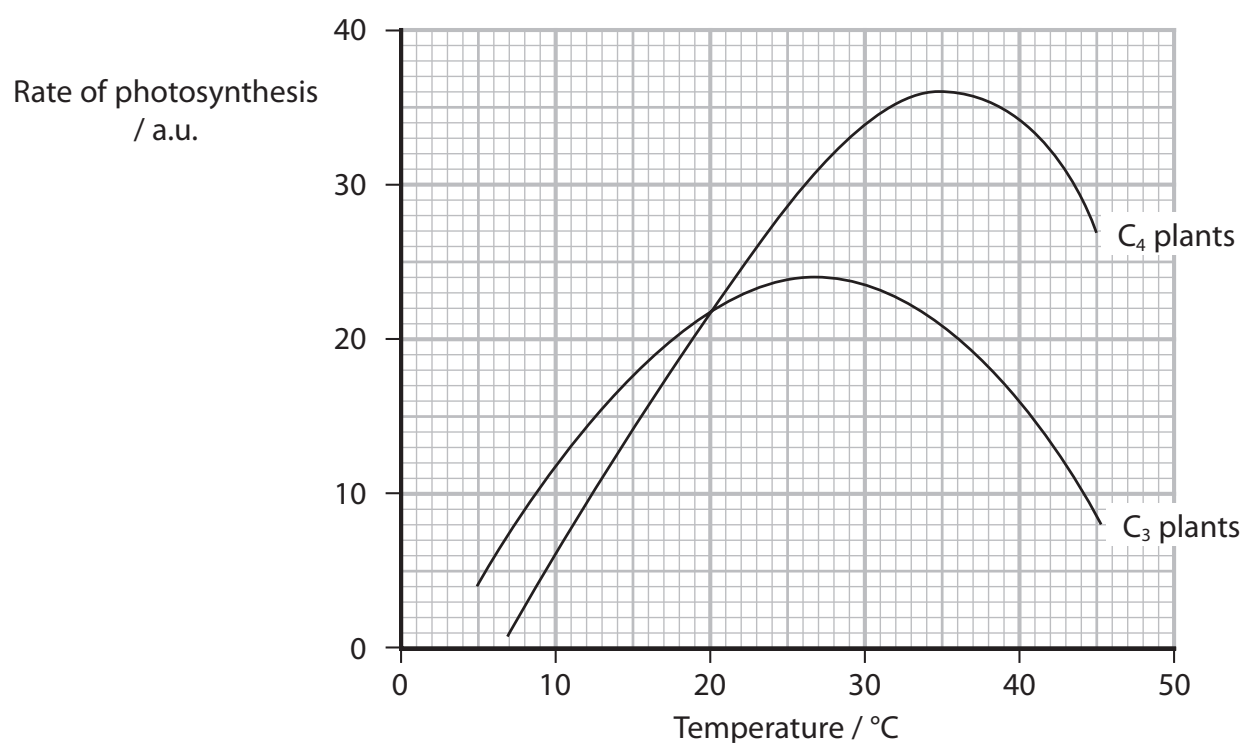
C

H

O

- (b) Two types of plant, C_3 and C_4 , have slightly different pathways of light-independent reactions.

The graph shows how the rate of photosynthesis is affected by temperature in these two types of plant.



- (i) Describe **three** conclusions that can be made about the differences in the effect of temperature on the rate of photosynthesis in these two types of plant.

(3)

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Calculate the Q_{10} for the enzymes in C_4 plants.

Use the formula:

$$Q_{10} = \frac{R_{t+10}}{R_t}$$

where R_t is the initial rate of reaction at 10°C .

Give your answer as a whole number.

(2)

Answer

(Total for Question 2 = 8 marks)